

Low-latency Edge AI with KubeEdge and OpenFAAS	Bringing intelligence closer to the edge is crucial for real-time applications in sectors like IoT and robotics. KubeEdge and OpenFAAS are open-source platforms enabling serverless deployments of AI models at the edge, empowering faster decision-making and minimized latency.
Decentralized Web with IPFS and Hyperledger Fabric	Centralized data servers raise privacy concerns and security risks. Projects like IPFS (InterPlanetary File System) and Hyperledger Fabric offer decentralized alternatives, distributing data across a network of nodes for a more resilient and secure web.
Conversational AI with Rasa and Rasa X	Building intelligent chatbots and virtual assistants requires powerful NLP tools. Rasa and Rasa X provide open-source frameworks for building and deploying conversational AI models, empowering developers to create engaging and interactive experiences.
Open Quantum Computing with OpenQCL and Cirq	The race to harness the power of quantum computing is hotting up. OpenQCL and Cirq are open-source libraries for developing and running quantum algorithms, making this exciting technology accessible to a wider audience.
Federated Learning with TensorFlow Federated and PySyft	Sharing knowledge without compromising data privacy is a critical challenge. Federated learning frameworks like TensorFlow Federated and PySyft enable collaborative training of AI models across multiple devices without sharing raw data, unlocking the potential for personalized and privacy-preserving AI.
Security Automation with OpenSCAP and Wazuh	With cyber threats constantly evolving, automating security responses is crucial. OpenSCAP and Wazuh provide open-source platforms for vulnerability scanning, intrusion detection, and incident response, empowering organizations to stay ahead of the curve.
Secure Containerization with gVisor and containerd	Containers are a popular way to package and deploy applications, but security concerns remain. gVisor and containerd offer secure containerization solutions, isolating applications from the host system and minimizing the attack surface.
Cloud-Native DevOps with ArgoCD and Kustomize	Managing deployments across multiple cloud environments can be complex. ArgoCD and Kustomize address this challenge, providing open-source tools for declarative configuration management and multi-cluster deployments, simplifying the cloud-native DevOps experience.
Open Data Science with JupyterLab and Apache Spark	Open data science platforms like JupyterLab and Apache Spark are vital for collaborative data analysis and machine learning. These projects provide users with interactive notebooks and distributed computing frameworks, empowering data scientists to tackle complex datasets efficiently.
Building Accessible Websites with WebAssembly and Blazor	WebAssembly is enabling new frontiers in web development, allowing execution of compiled languages like C++ and Rust in the browser. Frameworks like Blazor leverage WebAssembly to build interactive web applications and UIs with languages like C# and .NET, offering exciting possibilities for web development.

Figure 3: Open source tech trends in 2024

All in all, 2024 is going to be a year where development and adoption of AI, continued digitisation, and cloud computing will play a key role. Regardless of the technology, open source is playing a crucial role in accelerating its adoption. Industry leaders, including software development organisations, have realised the importance of open source to a level where they have started setting up open source program offices.

Open source program office (OSPO)

Yes, it's actually happening! The open source program office (OSPO) is officially a team with a leadership role like chief open source officer (COSO). Due to the speed at which open source innovation is happening, organisations are realising they cannot wait for professional software organisations to build products and services, especially companies that are interested in leading the

industry by pioneering solutions for tomorrow's opportunities. The OSPO is a cross-functional team responsible for strategising and implementing open source policies, solutions, and practices in an organisation. It helps to manage risks while maximising benefits associated with the use of open source software. It's a welcome sign that open source software is helping companies achieve higher growth, adding value beyond simply being free-to-use technology.

2024 promises to be an interesting year for all things technology, with AI-generated content, embedded monitoring across enterprise systems, and a renowned focus on reliability, sustainability, and security. Digitisation and the cloud will continue to be critical investments in enabling adoption of AI, IoT, 5G and quantum computing. As the year unfolds to new realities, the focus on efficiency will lead to further automation and an augmented digital workplace will take centre stage. **END** 

References

- Gartner Top Technology Trends for 2024; <https://www.gartner.com/en/articles/gartner-top-10-strategic-technology-trends-for-2024>
- Creating an Open Source Program Office (OSPO); <https://www.linuxfoundation.org/resources/open-source-guides/creating-an-open-source-program>

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Disclaimer: This article expresses the views of the author and not of the organisation he works in.